

Ronav Jaiswal

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EDUCATION

Carnegie Mellon University

B.S. in Computer Science and Artificial Intelligence (Graduate Coursework)

Aug 2024 - Dec 2027

GPA: 3.53/4.0

Courses: Distributed Systems, Database Systems, System Programming, Functional Programming, Deep Learning, Machine Learning, Computer Vision, Cybersecurity, Theoretical Computer Science, Probability Theory, Parallel Data Structures and Algorithms

SKILLS

Programming Languages: Python, C++, C, Java, SQL, JavaScript, TypeScript

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, REST APIs, Distributed Systems, Linux, Git

Backend & Databases: FastAPI, Node.js, PostgreSQL, ChromaDB, FAISS

Cloud, DevOps & Frameworks: AWS, Docker, Kubernetes, CI/CD, GitHub Actions, React, OpenCV, PyTorch, NumPy, Pandas

WORK EXPERIENCE

Wipro PARI

May 2026 - Aug 2026

AI and Robotics Intern, Software Controls

Auburn Hills, MI

- **Reduced vision deployment time by ~50%** by building an end-to-end computer vision platform in PyTorch and OpenCV.
- Built ViT-based defect detection pipelines across 10,000+ industrial images with automated training, validation, and deployment.
- Achieved **94% defect-detection accuracy** by fine-tuning pretrained ViT/DINOv2 representations with class-balanced training.
- **Increased profit margins by 3%** by deploying AI inspection software into PLC workflows for **\$10M+ Ford and Tesla projects**.

Herbochem Industries

June 2025 - Aug 2025

Software Engineering and AI Intern, Cloud and Security

New York City, NY

- Built a FastAPI document-processing backend over 10,000+ documents, **increasing retrieval throughput by 4x**.
- Developed scalable ingestion, embedding, vector-search, and retrieval pipelines using Python, FastAPI, and ChromaDB.
- **Improved document retrieval accuracy by 35%** by implementing reranking, MMR diversification, and vector search.
- Automated end-to-end compliance workflows using LangChain/LangGraph, **reducing manual processing effort by ~50%**.

Lee Lab@CMU: Machine Learning Department

Feb 2025 - Present

Undergraduate Research Fellow: Prof. Tai Sing Lee

Pittsburgh, PA

- Investigating memory and hierarchical cognitive control in world models, building on **novel inverse log-Z cortical magnification** for biologically inspired active vision.
- Developing hierarchical active-perception systems where Qwen-VLM reasons over sequential foveated observations and uncertainty dynamically controls fixation timing and location.
- Designing experiments on multi-timescale memory, latent belief updates, and coarse-to-fine reasoning in world models.

AWARDS AND RESEARCH PUBLICATIONS

TartanHacks Grand Prize - 2nd Place (CMU's Biggest Hackathon)

Jan 2026

- Built a real-time multimodal AI system combining MediaPipe pose estimation with Librosa/WebRTC audio analysis to detect breath timing and vocal strain. **Won 2nd Place at TartanHacks among 500+ teams** by fusing audio-visual predictions via IoU-based confidence aggregation for timestamped feedback.

Stylize Cluster Classify for Multidimensional Object Detection and Classification

IRJMETs

- **Published a novel Stylize-Cluster-Classify** computer vision pipeline combining image stylization, KMeans clustering, feature engineering, and dimensionality reduction for object classification.
- **Achieved 85.55% classification accuracy and a 13.79% improvement** over baseline ML models on 2,500+ images across 25 classes using cross-validation and hyperparameter tuning.

PROJECTS

HeapForge: High-Performance Dynamic Memory Allocator | C, Systems Programming, Memory Management

- Built a custom memory allocator implementing malloc, free, realloc, and calloc using mem_sbrk-based heap management.
- Achieved $\geq 74\%$ memory utilization while optimizing throughput through segregated free lists, block coalescing and footer elimination to reduce internal and external fragmentation.

HashGrid: Fault-Tolerant Distributed Compute System

- Engineered an exactly-once, in-order messaging protocol over lossy UDP with sliding-window flow control, epoch-based retransmission, exponential backoff, heartbeats, and checksum verification.
- Built a fault-tolerant distributed compute scheduler using goroutines and channels, dynamically load-balancing partitioned workloads across workers and automatically reassigning tasks after node failures.